

ABB Safety Configuration Report

A detailed description of functions and validation procedures can be found in the SafeMove application manual.

1. General Information.....	1
2. Safety Configuration.....	2
3. Safe I/O Configuration.....	6
4. Combinatorial Logic Configuration.....	8

1. General Information

Created by:	admin
Creation date:	2026-02-05T13:29:07.2247502+01:00
System name:	15000-500126
Configuration version:	3.04.00
Controller image version:	3.04.00
Checksum:	2A76 12BF 55B0 52FB 923A 205A DBCC A8FC 4018 30F3 4D61 FEB1 9E50 E21A 9057 649A
Checksum short:	E6710D10

2. Safety Configuration

2.1 Drive Modules

Drive Module 1 Configuration

Max speed manual mode
250.000 mm/s

Drive Module 1 Configuration - ROB_1

Safe brake ramp start speed offset	Elbow offset	Baseframe
100.000 mm/s	x: -50.000 mm y: -50.000 mm z: 50.000 mm	Position x: 0.000 mm y: 0.000 mm z: 0.000 mm Orientation x: 0.000 deg y: 0.000 deg z: 0.000 deg

ROB_1 - Upper Arm Geometries

Upper Arm Geometries - Robot_Capsule_1

Type	Radius	Start	End
Capsule	160.000 mm	x: -30.356 mm y: -22.120 mm z: 30.485 mm	x: 186.565 mm y: 0.000 mm z: 100.000 mm

Upper Arm Geometries - Robot_Capsule_2

Type	Radius	Start	End
Capsule	140.000 mm	x: 380.000 mm y: 30.000 mm z: 150.000 mm	x: 520.000 mm y: 10.000 mm z: 150.000 mm

ROB_1 Verified: _____

Drive Module 1 Configuration - Synchronization

Activation	Synchronization status
Software synchronization	No signal

Synchronization - Sync position

Joint	position
1	0.000 deg
2	0.000 deg
3	0.000 deg
4	0.000 deg
5	0.000 deg
6	0.000 deg

Synchronization Verified: _____

Drive Module 1 Configuration - Cyclic Brake Check

Warning only, no stop	Max CBC test interval	Pre warning time	Standstill tolerance	Supervision threshold
true	200 h	6 h	2 rad (motor side)	0.02 rad (motor side)

2. Safety Configuration

Cyclic Brake Check - Joints

Joint	Enabled
1	false
2	false
3	false
4	false
5	false
6	false

Drive Module 1 Configuration - Tools

Tools - Tool

Activation	Active status	TCP	Orientation	Rigid Body
Permanently active	No signal	x: 0.000 mm y: 0.000 mm z: 0.000 mm	x: 0.000 deg y: 0.000 deg z: 0.000 deg	Mass: 0.000 Kg x: 0.000 mm y: 0.000 mm z: 0.000 mm ix: 0.000 Kgm ² iy: 0.000 Kgm ² iz: 0.000 Kgm ²

Tool - Speed Supervision Points (Flange Coordinates)

Number	X	Y	Z
--------	---	---	---

Tool - Tool Geometries

Tool Geometries - Sphere

Type	Radius	Center
Sphere	50.000 mm	x: 0.000 mm y: 0.000 mm z: 0.000 mm

Tool Geometries Verified: _____

Tool Verified: _____

Drive Module 1 Configuration - Safe Zones

Safe Zones - Gesamtzone

Tool speed supervision priority
Normal

Gesamtzone - Coordinates

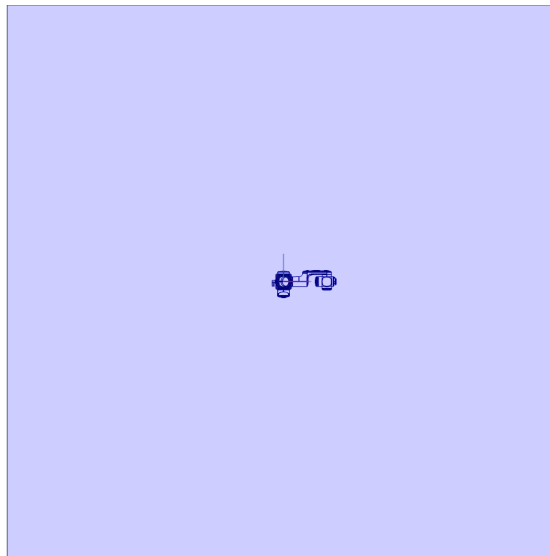
Top	Bottom
3000.000 mm	-3000.000 mm

Vertices

Number	X	Y
1	-3000.000 mm	-3000.000 mm
2	3000.000 mm	-3000.000 mm
3	3000.000 mm	3000.000 mm
4	-3000.000 mm	3000.000 mm

2. Safety Configuration

Gesamtzone - Floor plan



Gesamtzone - Tool Speed Supervisions

Tool Speed Supervisions - Gesamtzone_TSP

Activation	Function active status	Violation stop category	Violation signal
Permanently active	No signal	Category1Stop	No signal
Max speed			
500.000 mm/s			

Gesamtzone Verified: _____

Drive Module 1 Configuration - Safe Ranges

Safe Ranges - Achsbegrenzung

Achsbegrenzung - Joint limits

Joint	Lower bound	Upper bound	Invert
1	-180.000 deg	180.000 deg	false
2	-90.000 deg	90.000 deg	false
3	-225.000 deg	85.000 deg	false
4	-180.000 deg	180.000 deg	false
5	-180.000 deg	180.000 deg	false
6	-270.000 deg	270.000 deg	false

Safe Ranges Verified: _____

2.2 Stop Configurations

ExternalEmergencyStop

Stop category	Decoupled	
Category1Stop	Coupled	

ExternalEmergencyStop Verified: _____

2. Safety Configuration

InternalEmergencyStop

Stop category	
Category1Stop	

InternalEmergencyStop Verified: _____

ProtectiveStop

Mode	Stop category	
Auto	Category1Stop	

ProtectiveStop Verified: _____

3. Safe I/O Configuration

3.1 Global Signals

Name	Type	Default	Offset	Width	Direction	Protected
AutomaticMode	BOOL	0	0	1	output	false
bAlex	BOOL	1	1	1	output	false
DriveEnable	BOOL	0	2	1	output	false
DriveEnableFeedback	BOOL	0	3	1	output	false
EmergencyStopActivated	BOOL	0	4	1	output	false
EnableSwitch	BOOL	0	5	1	output	false
ExternalEmergencyStopStatus	BOOL	0	6	1	output	false
LocalEmergencyStopStatus	BOOL	0	7	1	output	false
ManualFullSpeedMode	BOOL	0	8	1	output	false
ManualMode	BOOL	0	9	1	output	false
ProtectiveStop	BOOL	0	10	1	output	false
SafetyEnable	BOOL	1	11	1	output	false

3.2 Networks

Feedback

Feedback - Devices

SC_Feedback_Dev

Devices - Signals

Devices - Output

Name	Type	Default	Offset	Width	Direction	Protected
AutomaticMode	BOOL	0	0	1	output	false
bAlex	BOOL	1	1	1	output	false
DriveEnable	BOOL	0	2	1	output	false
DriveEnableFeedback	BOOL	0	3	1	output	false
EmergencyStopActivated	BOOL	0	4	1	output	false
EnableSwitch	BOOL	0	5	1	output	false
ExternalEmergencyStopStatus	BOOL	0	6	1	output	false
LocalEmergencyStopStatus	BOOL	0	7	1	output	false
ManualFullSpeedMode	BOOL	0	8	1	output	false
ManualMode	BOOL	0	9	1	output	false
ProtectiveStop	BOOL	0	10	1	output	false
SafetyEnable	BOOL	1	11	1	output	false

3.3 Function Mappings

Function	Signal	Mandatory	Description	Group
AutomaticMode	AutomaticMode	true		
DriveEnable	DriveEnable	true		
DriveEnableFeedback	DriveEnableFeedback	true		
EmergencyStopActivated	EmergencyStopActivated	true		
EnableSwitch	EnableSwitch	true		

3. Safe I/O Configuration

Function	Signal	Mandatory	Description	Group
ExternalEmergencyStopStatus	ExternalEmergencyStopStatus	true		
LocalEmergencyStopStatus	LocalEmergencyStopStatus	true		
ManualFullSpeedMode	ManualFullSpeedMode	true		
ManualMode	ManualMode	true		
ProtectiveStop	ProtectiveStop	true		
SafetyEnable	SafetyEnable	true		

4. Combinatorial Logic Configuration

4.1 Pre Logic

Name	Expression
------	------------

4.2 Post Logic

Name	Expression
------	------------

Complete functionality verified and tested

Date

Signature